Lab 7: Fuzzing for Input Validation Bugs

Name: **Nikhil Suhas Joshi** SRN: **PES1UG24AM441**

Step 2: Run the buggy processor.py

`python processor.py` (inputs: 2TB, 1024:0, HOT_TIER). The original crashes with ZeroDivisionError.

```
==== Running parse_storage_quota ====
Enter quota (e.g. 500MB, 10GB, 2TB): Quota: 2000.00 GB

==== Running compute_compression_ratio ====
Enter raw:compressed bytes (e.g. 1024:256): Traceback (most recent call last):
  File "original/processor.py", line 56, in <module>
    main()
  File "original/processor.py", line 49, in main
    print("Compression Ratio:", compute_compression_ratio(c_input))
                                ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "original/processor.py", line 26, in compute_compression_ratio
    ratio = raw_bytes / comp_bytes
            ~~~~~~~~~~^~~~~~~~~~~~
ZeroDivisionError: float division by zero
```

Step 4: Run the Hypothesis tests on the buggy code

`python -m pytest test_processor.py`. All 3 fuzz tests FAIL (output trimmed to the final part).

```
  |     def test_lookup_bandwidth_tier_rate_safe(s):
  |                    ^^^
  | BaseExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "original/test_processor.py", line 38, in test_lookup_bandwidth_tier_rate_safe
    |     lookup_bandwidth_tier_rate(s)
    |   File "original/processor.py", line 40, in lookup_bandwidth_tier_rate
    |     return tier_rates[prefix]
    |            ~~~~~~~~~~^^^^^^^^
    | KeyError: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "original/test_processor.py", line 42, in test_lookup_bandwidth_tier_rate_safe
    |     pytest.fail(f"lookup_bandwidth_tier_rate({s!r}) crashed with {type(e).__name__}: {e}")
    | Failed: lookup_bandwidth_tier_rate('') crashed with KeyError: ''
    | Failing test case: test_lookup_bandwidth_tier_rate_safe(
    |     s='',  # or any other generated value
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "original/test_processor.py", line 38, in test_lookup_bandwidth_tier_rate_safe
    |     lookup_bandwidth_tier_rate(s)
    |   File "original/processor.py", line 34, in lookup_bandwidth_tier_rate
    |     prefix = tier_descriptor.strip().upper()[:3]
    |              ^^^^^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'strip'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "original/test_processor.py", line 42, in test_lookup_bandwidth_tier_rate_safe
    |     pytest.fail(f"lookup_bandwidth_tier_rate({s!r}) crashed with {type(e).__name__}: {e}")
    | Failed: lookup_bandwidth_tier_rate(None) crashed with AttributeError: 'NoneType' object has no attribute 'strip'
    | Failing test case: test_lookup_bandwidth_tier_rate_safe(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
========================== short test summary info ==========================
FAILED test_processor.py::test_parse_storage_quota_no_crash - Failed: parse_s...
FAILED test_processor.py::test_compute_compression_ratio_safe - Failed: compu...
FAILED test_processor.py::test_lookup_bandwidth_tier_rate_safe - BaseExceptio...
============================ 3 failed in 1.42s ==============================
```

Step 5: Run the fixed processor.py

`python processor.py` (same inputs). The bad input is rejected cleanly instead of crashing.

```
==== Running parse_storage_quota ====
Enter quota (e.g. 500MB, 10GB, 2TB): Quota: 2000.00 GB

==== Running compute_compression_ratio ====
Enter raw:compressed bytes (e.g. 1024:256): Error: raw must be >= 0 and compressed must be > 0

==== Running lookup_bandwidth_tier_rate ====
Enter tier descriptor (e.g. HOT_TIER, COLD, ARCHIVE): Rate per GB: $ 0.08
```

Step 6: Re-run the tests against the fixed processor.py

`python -m pytest test_processor.py -v`. All 3 tests PASS.

```
============================= test session starts ==============================
platform linux -- Python 3.12.3, pytest-9.2.0, pluggy-1.7.0 -- /usr/bin/python3
hypothesis profile 'default'
rootdir: /tmp/L6_L7_12_cloud_storage_tier-main/Lab 7
plugins: platformdirs-4.13.0, hypothesis-6.170.15, cov-7.1.0
collecting ... collected 3 items

test_processor.py::test_parse_storage_quota_no_crash PASSED              [ 33%]
test_processor.py::test_compute_compression_ratio_safe PASSED            [ 66%]
test_processor.py::test_lookup_bandwidth_tier_rate_safe PASSED           [100%]

============================== 3 passed in 1.25s ===============================
```